

PM and Alarm Collection

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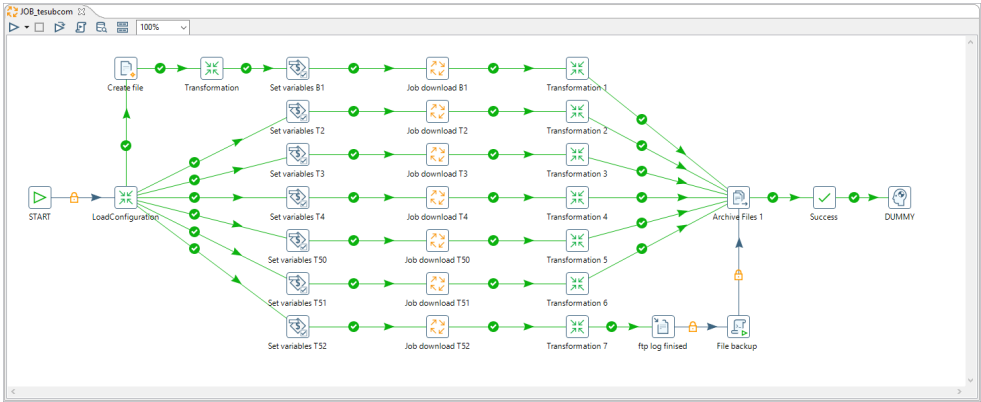
TEMS

PM data collection

Mediation application

Pentaho Data Integration(PDI) application collects PM data files through FTP from the TEMS OCH server and extracts the Key Performance Index(KPI) values and stores it in the database(MariaDB).

PDI kettle job(JOB_tesubcom.kjb) which starts the application



Environment details

	App Server	Application path	TEMS FTP Server	Backup disk	Git	Collection time	Collection frequency
Production	10.7.2.104	pcsunavailableSeconds/hawaiki_il/tesubcom	172.29.135.30 (AUSY_A)	10.7.2.190:/mediation/tesubcom	https://gitlab.local.hawaiki.co.nz/hawaiki-dev/kettle/tree/master/tesubcom	02:00 UTC	24 hours
Pre-production	10.10.12.108		172.29.136.30 (NZHQ)	10.10.13.201:/mediation/tesubcom		06:00 UTC	24 hours

PM collection application details

File	Details
JOB_tesubcom.kjb	This job handles everything from start to finish of the application. All other jobs and transformation are called from this job.
LoadConfiguration.ktr	Loads configuration values from 'config.properties' file.
InitiateFTPlogs.ktr	Creates the log file 'ftp_logs.txt' inside '/hawaiki_ll/tesubcom/ftp_logs' folder.
DownloadFile.kjb	Downloads the file from the TEMS OCH server.
LoadTEMSFiles.ktr	Extracts PM details from the downloaded file and writes to the table(tems_och).
config.properties	The configuration file contains FTP and database details.
script/test-script.sh	Moves the downloaded file to the backup disk.
ftp_logs/ftp_logs.txt	This log file is sent in the email report.
tems_och.sh	A shell script that runs the PM collection application kettle job. This script is called from the mediation servers crontab.

Email report application details

File	Details
JOB_sendEmail.kjb	This job handles everything from start to finish of the application. All other jobs and transformation are called from this job.
LoadConfiguration.ktr	Loads configuration values from 'config.properties' file.
GenerateMissingPMs.ktr	Finds the completely missing PM sources from the database tems_och and writes it to the log file 'ftp_logs/ftp_logs.txt'.
GenerateFTPlogs.ktr	Finds the partially missing PM sources from the database tems_och and writes it to the log file 'ftp_logs/ftp_logs.txt'.
SendEmail.ktr	Reads the log file and sends the email.
config.properties	The configuration file which contains mail server details, sender and destination email address, email subject.
ftp_logs/ftp_logs.txt	This log file is sent as an email report.
tems_email.sh	A shell script that runs the email report kettle job. This script is called from the mediation servers crontab.

Database

MariaDB details

	MariaDB Server	Schema	Tables used	Database backup
Production	ausy-db-mdb-oss-m2.local.hawaiki.co.nz	pm_db	service_source, tems_och	
Pre-production	10.10.13.201			

Tables

service_source	tems_och
id INT(11)	id INT(10)
customer VARCHAR(16)	Date/Time DATETIME
customer_sid VARCHAR(16)	customer_sid VARCHAR(24)
pm_source VARCHAR(128)	PM Source VARCHAR(256)
visible_to_customer TINYINT(4)	node_id VARCHAR(64)
route_end VARCHAR(3)	Type TINYTEXT
segment_end VARCHAR(3)	Q (dB) Minimum DOUBLE
source VARCHAR(16)	Q (dB) Maximum DOUBLE
node_id VARCHAR(64)	Q (dB) Average DOUBLE
to_from VARCHAR(2)	Q (dB) Std Dev DOUBLE
start_date DATETIME	Link Fail Seconds - In DOUBLE
end_date DATETIME	Physical Error Count - In DOUBLE
	Frame Check Sequence Error Count - In DOUBLE
	Link Fail Seconds - Out DOUBLE
	Physical Error Count - Out DOUBLE
	Frame Check Sequence Error Count - Out DOUBLE
	Unavailable Seconds (JAS) DOUBLE
	Number of Seconds in Bin - Tx Line Card DOUBLE
	Number of Seconds in Bin - Rx Line Card DOUBLE

Grafana Stats portal

PM data are monitored using Grafana dashboards.



	Dashboard	URL	Git
Production - Customer	Customer name + " Service 8X Factors"	https://stats.local.hawaiki.co.nz/stats/d/MOM9mKik/amazon-service-8x-factors?orgId=1	https://gitlab.local.hawaiki.co.nz/hawaiki-dev/kettle/tree/master/grafana
Production - NOC	Customer Service 8X Factors	https://stats-new.local.hawaiki.co.nz/stats/d/pVmmB4dmz/customer-service-8x-factors?orgId=1	
Pre-production	Customer Service 8X Factors	http://stats.test.local.hawaiki.co.nz/d/pVmmB4dmz/customer-service-8x-factors?orgId=1	

As of September 2019, we have Q Factor only for TEMS, Ciena Q Factor is yet to implement.

For Q Factor Environment details, please refer to PM Data Collection → Mediation application → Environment details

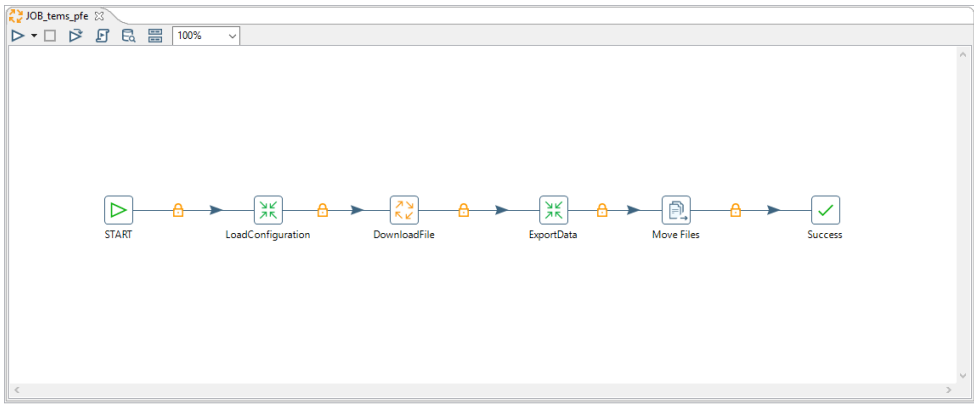
File	Details
JOB_sendQFactorReportEmail.kjb	This job handles everything from start to finish of the application. All other jobs and transformation are called from this job.
LoadConfiguration.ktr	Loads configuration values from ' <i>config.properties</i> ' file.
InitiateQFactorlogs.ktr	Initiate Q Factor log file ' <i>email/q_factor_report.txt</i> '.
GenerateFTPlogs.ktr	Reads Q Factor values from the table tems_och and writes it to the log file ' <i>email/q_factor_report.txt</i> '.
SendQFactorReportEmail.ktr	Reads the log file and sends the email.
config.properties	The configuration file which contains mail server details, sender and destination email address, email subject.
email/q_factor_report.txt	This log file is sent as an email report.
tems_email.sh	A shell script that runs the email report kettle job. This script is called from the mediation servers crontab.

Please refer to PM Data Collection → Database

[illegible]

	Dashboard	URL	Git
Production - NOC	Q Factor	https://stats-new.local.hawaiki.co.nz/stats/d/wCyooRdiz/q-factor?orgId=1	https://gitlab.local.hawaiki.co.nz/hawaiki-dev/kettle/tree/master/grafana
Pre-production	Q Factor	http://stats.test.local.hawaiki.co.nz/d/wCyooRdiz/q-factor?orgId=1	

PDI kettle job(JobTemsPfe.kjb) which starts the application



Environment details

	App Server	Application path	Git	Collection time	Collection frequency
Production	10.7.2.104	/hawaiki_i/tesubcom-pfe	https://gitlab.local.hawaiki.co.nz/hawaiki-dev/kettle/tree/test/tems-pfe	06:29 UTC	24 hours
Pre-production	10.10.12.108			06:29 UTC	24 hours

PFE collection application details

File	Details
JobTemsPfe.kjb	This job handles everything from start to finish of the application. All other jobs and transformation are called from this job.
LoadConfiguration.ktr	Loads configuration values from 'config.properties' file.
DownloadFile.kjb	Downloads the file from the TEMS server.
ExportData.ktr	Extracts PFE details from the downloaded file and writes to the table(tems_pfe).
config.properties	The configuration file contains FTP and database details.
tems_pfe.sh	A shell script that runs the PFE collection application kettle job. This script is called from the mediation servers crontab.
tems_pfe.log	Application log.

Database

MariaDB details

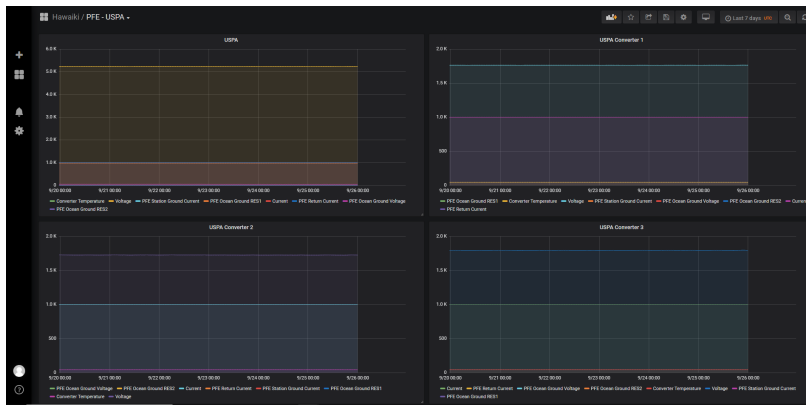
	MariaDB Server	Schema	Tables used	Database backup
Production	ausy-db-mdb-oss-m2.local.hawaiki.co.nz	pm_db	tems_pfe	
Pre-production	10.10.13.201			

Tables

tems_pfe
id INT (11)
suspect TINYTEXT
Date/Time DATETIME
PM Source TINYTEXT
Type TINYTEXT
Current (mA) Minimum DOUBLE
Current (mA) Maximum DOUBLE
Current (mA) Average DOUBLE
Voltage (V) Minimum DOUBLE
Voltage (V) Maximum DOUBLE
Voltage (V) Average DOUBLE
PFE Return Current (mA) Minimum DOUBLE
PFE Return Current (mA) Maximum DOUBLE
PFE Return Current (mA) Average DOUBLE
PFE Station Ground Current (mA) Minimum DOUBLE
PFE Station Ground Current (mA) Maximum DOUBLE
PFE Station Ground Current (mA) Average DOUBLE
PFE Ocean Ground Voltage (V) Minimum DOUBLE
PFE Ocean Ground Voltage (V) Maximum DOUBLE
PFE Ocean Ground Voltage (V) Average DOUBLE
PFE Ocean Ground RES 1 (mA) Minimum DOUBLE
PFE Ocean Ground RES 1 (mA) Maximum DOUBLE
PFE Ocean Ground RES 1 (mA) Average DOUBLE
PFE Ocean Ground RES 2 (mA) Minimum DOUBLE
PFE Ocean Ground RES 2 (mA) Maximum DOUBLE
PFE Ocean Ground RES 2 (mA) Average DOUBLE
Converter Temperature (C) Minimum DOUBLE
Converter Temperature (C) Maximum DOUBLE
Converter Temperature (C) Average DOUBLE

Grafana Stats portal

PFE details are monitored using Grafana dashboards.



Grafana portal details

	Dashboard	URL	Git
Production - NOC	PFE - ASGO	ASGO: https://stats-new.local.hawaiki.co.nz/stats/d/FyuxBwfik/pfe-asgo?orgId=1	https://gitlab.local.hawaiki.co.nz/hawaiki-dev/kettle/tree/master/grafana
	PFE - AUSY	AUSY: https://stats-new.local.hawaiki.co.nz/stats/d/dFecmABmk/pfe-ausy?orgId=1	
	PFE - HIKA	HIKA: https://stats-new.local.hawaiki.co.nz/stats/d/nipeEwfmk/pfe-hika?orgId=1	
	PFE - NZMA	NZMA: https://stats-new.local.hawaiki.co.nz/stats/d/yhhPrsfmk/pfe-nzma?orgId=1	
	PFE - USPA	USPA: https://stats-new.local.hawaiki.co.nz/stats/d/HYkpLQfiz/pfe-uspa?orgId=1	
Pre-production	PFE - ASGO	ASGO: http://stats.test.local.hawaiki.co.nz/d/FyuxBwfik/pfe-asgo?orgId=1	
	PFE - AUSY	AUSY: http://stats.test.local.hawaiki.co.nz/d/dFecmABmk/pfe-ausy?orgId=1	
	PFE - HIKA	HIKA: http://stats.test.local.hawaiki.co.nz/d/nipeEwfmk/pfe-hika?orgId=1	
	PFE - NZMA	NZMA: http://stats.test.local.hawaiki.co.nz/d/yhhPrsfmk/pfe-nzma?orgId=1	
	PFE - USPA	USPA: http://stats.test.local.hawaiki.co.nz/d/HYkpLQfiz/pfe-uspa?orgId=1	

Alarm collection - Bash Script Application

TEMS alarm collection is written in bash script which collects alarms through SNMP Walk.

Mediation application

Alarms received are cross-checked with customer assigned ports and sends to OpenNMS.

All the alarms which are assigned to a customer are forwarded to Impacted Services in OpenNMS.

All the alarms which are not assigned to a customer are forwarded to Subcom Alarm in OpenNMS.

Environment details

	App Server	Application path	TEMS Server	Git	Collection frequency
Production	10.7.1.41	/opt/HawaikiTools/TemsAlarms	activetems.local.hawaiki.co.nz	https://gitlab.local.hawaiki.co.nz/hawaiki-dev/kettle/tree/master/tems-alarms	1 minute

Tems alarm collection application details

File	Details
TemsAlarms.sh	Bash script which contains application code.
TemsCustomerPms.csv	File which contains customer assigned port details. (This file has to be updated when a new service is added for PM collection)
TemsAlarmsChecksum.csv	A temporary file used for filtering duplicate alarms.
NewAlarmIDs.csv	A temporary file used for storing new alarm details.
alarms-timestamp.txt	A temporary file used by the script.
alarms-type.txt	A temporary file used by the script.
alarms-cause.txt	A temporary file used by the script.
alarms-severity.txt	A temporary file used by the script.
alarms-source.txt	A temporary file used by the script.

Email notifications

Condition	to address
alarms match customer assigned port	itdev@hawaikicable.co.nz operations-noc@hawaikicable.co.nz

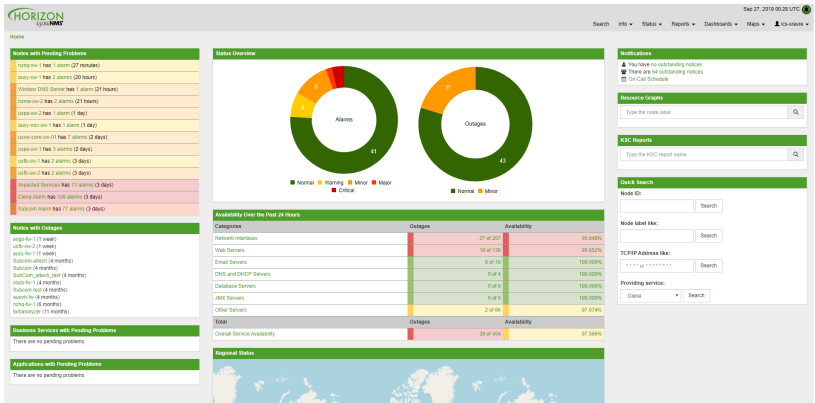
Database

TEMS alarms are not stored in the database by mediation application.

OpenNMS application stores it in PostgreSQL.

OpenNMS

TEMS alarms are monitored using OpenNMS



OpenNMS portal details

	URL
Production	https://opennms.local.hawaiki.co.nz/opennms/index.jsp

Ciena

Ciena MCP Blueplanet servers

MCP cluster details

Site	URL	IP
USFB	http://mcpcciena.local.hawaiki.co.nz	
AUSY	http://mcpcciena2.local.hawaiki.co.nz	

Changing PM collection location

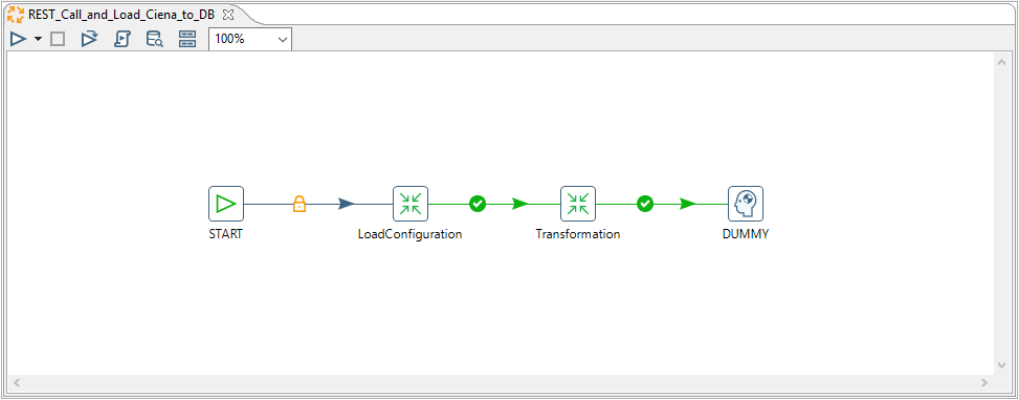
Token database table (ciena_tokens) should be cleared after changing the PM collection location.

PM collection - 6500

Mediation application

Pentaho Data Integration(PDI) application collects PM data by calling REST API's to CIENA MCP server and extracts the Key Performance Index(KPI) values and stores it in the database(MariaDB).

PDI kettle job(REST_Call_and_Load_Ciena_to_DB.kjb) which starts the application



Environment details

	App Server	Application path	Ciena REST Server	Git	Collection time	Collection frequency
Production	10.7.2.104	/hawaiki_ill/ciena	mcpcciena2.local.hawaiki.co.nz (AUSY)	https://gitlab.local.hawaiki.co.nz/hawaiki-dev/kettle/tree/master/ciena	07, 23, 37, 51	15 minutes
Pre-production	10.10.12.108				06, 18, 37, 48	15 minutes

REST Request

Request body	Mapped Parameters	KPI's collected	API
{	Link Fail Seconds - In → UAS-E (RCV)	Link Fail Seconds - In,	http://mcpcciena2.local.hawaiki.co.nz/pm/api/v1/queries

Request body	Mapped Parameters	KPI's collected	API
<pre> "data": { "attributes": { "criteria": { "binIndex": "1", "granularity": ["15_MINUTE"], "includeThreshold": false, "parameters": { "UAS-E", "FCSERR-E", "INFRAMESERR-E", "OUTFRAMESERR-E","ES-E","SES-E"], "suppress": "NONE" }, "measurementPoints": [{ "nodeId": "\$NCID", "resourceId": "\$NCID:TPE_\$NCID::EQPT_\$SHELF_\$SLOT_\$PORT-PTP", "resourceObject": { "location": { "port": "\$PORT", "shelf": "\$SHELF", "slot": "\$SLOT" }, "type": "pmData" }, "resourceType": "TPE" }] } } </pre>	Link Fail Seconds - Out → UAS-E (TRMT) Physical Error Count - In → INFRAMESERR-E(RCV) Physical Error Count - Out → OUTFRAMESERR-E(TRMT) Frame Check Sequence Error Count - In → FCSERR-E(RCV) Frame Check Sequence Error Count - Out → FCSERR-E(TRMT) Errored Seconds - In → ES-E (RCV) Errored Seconds - Out → ES-E (TRMT) Severely Errored Seconds - In → SES-E (RCV) Severely Errored Seconds - Out → SES-E (TRMT) Number of Seconds in Bin - Tx Line Card → Based On Time Number of Seconds in Bin - Rx Line Card → Based On Time	Link Fail Seconds - Out, Physical Error Count - In, Physical Error Count - Out, Frame Check Sequence Error Count - In, Frame Check Sequence Error Count - Out, Errored Seconds - In, Errored Seconds - Out, Severely Errored Seconds - In, Severely Errored Seconds - Out, Number of Seconds in Bin - Tx Line Card, Number of Seconds in Bin - Rx Line Card	

PM collection application details

File	Details
REST_Call_and_Load_Ciena_to_DB.kjb	This job handles everything from start to finish of the application. All other jobs and transformation are called from this job.
ciena_load_configuration.ktr	Loads configuration values from ' <i>config.properties</i> ' file.
ciena_report_extraction.ktr	Handles getting access token and pm extraction
ciena_token.ktr	Retrieves access token from Ciena MCP using REST API.
ciena_pm.ktr	Collects PM details from Ciena MCP using REST API and stores the data into the ' <i>tems_och</i> ' table.
config.properties	The configuration file contains the Ciena server and database details.
ciena.log	Application log file.
ciena_och.sh	A shell script that runs the PM collection application kettle job. This script is called from the mediation servers crontab.

Email report application details

File	Details
JOB_sendEmail.kjb	This job handles everything from start to finish of the application. All other jobs and transformation are called from this job.
ciena_load_configuration.ktr	Loads configuration values from ' <i>config.properties</i> ' file.
initiate_pm_missing_logs.ktr	create log file email/email-body.txt
generate_missing_pms.ktr	writes missing PM details to email/email-body.txt
send_email.ktr	Reads the log file and sends the email.
config.properties	The configuration file which contains mail server details, sender and destination email address, email subject.
email/email-body.txt	This file's content is sent as an email report.
send_email.log	Application log file.
send_email.sh	A shell script that runs the email report kettle job. This script is called from the mediation servers crontab.

Database

MariaDB details

	MariaDB Server	Schema	Tables used	Database backup
Production	ausy-db-mdb-oss-m2.local.hawaki.co.nz	pm_db	ciena_tokens, service_source, tems_och	
Pre-production	10.10.13.201			

Tables

service_source	tems_och
id INT(11)	id INT(10)
customer VARCHAR(16)	Date/Time DATETIME
customer_sid VARCHAR(16)	customer_sid VARCHAR(24)
pm_source VARCHAR(128)	PM Source VARCHAR(256)
visible_to_customer TINYINT(4)	node_id VARCHAR(64)
route_end VARCHAR(3)	Type TINYTEXT
segment_end VARCHAR(3)	Q (dB) Minimum DOUBLE
source VARCHAR(16)	Q (dB) Maximum DOUBLE
node_id VARCHAR(64)	Q (dB) Average DOUBLE
to_from VARCHAR(2)	Q (dB) Std Dev DOUBLE
start_date DATETIME	Link Fail Seconds - In DOUBLE
end_date DATETIME	Physical Error Count - In DOUBLE
	Frame Check Sequence Error Count - In DOUBLE
	Link Fail Seconds - Out DOUBLE
	Physical Error Count - Out DOUBLE
	Frame Check Sequence Error Count - Out DOUBLE
	Unavailable Seconds (JAS) DOUBLE
	Number of Seconds in Bin - Tx Line Card DOUBLE
	Number of Seconds in Bin - Rx Line Card DOUBLE

Grafana Stats portal

PM data are monitored using Grafana dashboards.



Grafana portal details

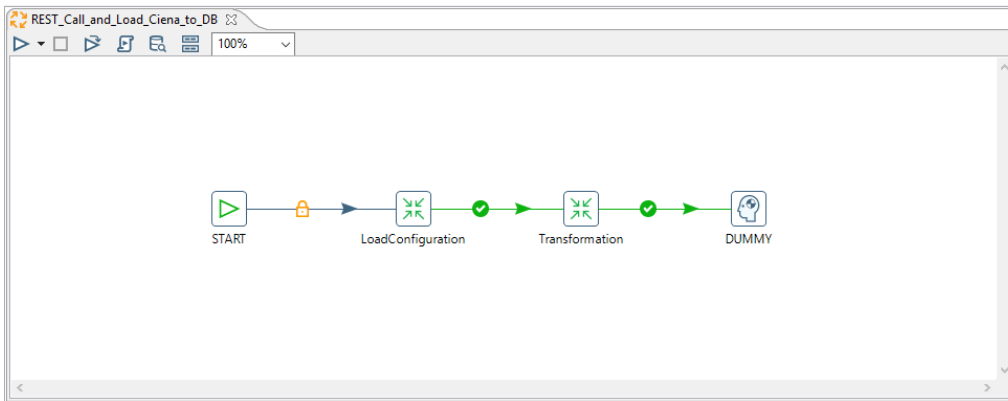
	Dashboard	URL	Git
Production - Customer	Customer name + " Service 8X Factors"	https://stats.local.hawaiki.co.nz/stats/d/ukwwFgPiz/hawaiian-tel-service-8x-factors?orgId=1&var-customer=HAWAIIANTEL&var-service_id=OMA-BX0035&var-pm_source=All	https://gitlab.local.hawaiki.co.nz/hawaiki-dev/kettle/tree/master/grafana
Production - NOC	Customer Service 8X Factors	https://stats-new.local.hawaiki.co.nz/stats/d/pVMmB4dmz/customer-service-8x-factors?orgId=1&var-customer=HAWAIIANTEL&var-service_id=OMA-BX0035&var-pm_source=All	
Pre-production	Customer Service 8X Factors	http://stats.test.local.hawaiki.co.nz/d/ZUpo58VWk/zayo-service-8x-factors?orgId=1	

PM collection - OTU

Mediation application

Pentaho Data Integration(PDI) application collects PM data by calling REST API's to CIENA MCP server and extracts the Key Performance Index(KPI) values and stores it in the database(MariaDB).

PDI kettle job(REST_Call_and_Load_Ciena_to_DB.kjb) which starts the application



Environment details

	App Server	Application path	Ciena REST Server	Git	Collection time	Collection frequency
Production	10.7.2.104	/hawaiki_jl/ciena-2	mcp-ciena2.local.hawaiki.co.nz (AUSY)	https://gitlab.local.hawaiki.co.nz/hawaiki-dev/kettle/tree/master/ciena	06, 21, 36, 51	15 minutes
Pre-production	10.10.12.108				06, 21, 36, 51	15 minutes

REST Request

Request body	Mapped Parameters	KPI's collected	API
<pre>{ "data": { "attributes": { "criteria": { "binIndex": "1", "granularity": ["15_MINUTE"], "locationNative": "NEND" }, "measurementPoints": [{ "model": "\$NCID", "resourceId": "\$NCID::TPE_\$NCID::EQPT_\$SHELF_\$SLOT_\$PORT-PTP", "resourceObject": { "location": { "port": "\$PORT", "shelf": "\$SHELF", "slot": "\$SLOT" }, "type": "pmData" }, "resourceType": "TPE" }], "type": "queries" } }</pre>	Link Fail Seconds - In → UAS-E (RCV) Physical Error Count - In → INFRAMESERR-E(RCV) Frame Check Sequence Error Count - In → FCSERR-E(RCV)	Link Fail Seconds - In, Physical Error Count - In, Frame Check Sequence Error Count - In,	http://mcpcciena2.local.hawaiki.co.nz/pm/api/v1/queries

PM collection application details

File	Details
REST_Call_and_Load_Ciena_to_DB.kjb	This job handles everything from start to finish of the application. All other jobs and transformation are called from this job.
ciena_load_configuration.ktr	Loads configuration values from 'config.properties' file.
ciena_report_extraction.ktr	Handles getting access token and pm extraction
ciena_token.ktr	Retrieves access token from Ciena MCP using REST API.
ciena_pm.ktr	Collects PM details from Ciena MCP using REST API and stores the data into the 'tems_och' table.
config.properties	The configuration file contains the Ciena server and database details.
ciena.log	Application log file.
ciena_och.sh	A shell script that runs the PM collection application kettle job. This script is called from the mediation servers crontab.

Database

MariaDB details

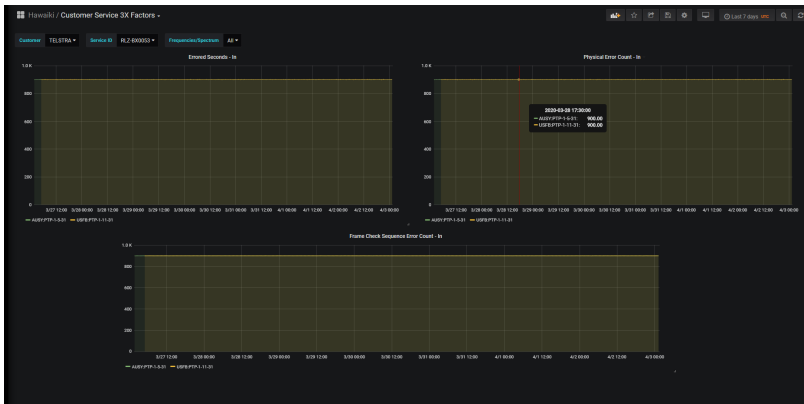
	MariaDB Server	Schema	Tables used	Database backup
Production	ausy-db-mdb-oss-m2.local.hawaiki.co.nz	pm_db	ciena_tokens, service_source, tems_och	
Pre-production	10.10.13.201			

Tables

service_source	tems_och
- id INT(11) - customer VARCHAR(16) - customer_sid VARCHAR(16) - pm_source VARCHAR(128) - visible_to_customer TINYINT(4) - route_end VARCHAR(3) - segment_end VARCHAR(3) - source VARCHAR(16) - node_id VARCHAR(64) - to_from VARCHAR(2) - start_date DATETIME - end_date DATETIME	- id INT(10) - Date/Time DATETIME - customer_sid VARCHAR(24) - PM Source VARCHAR(256) - node_id VARCHAR(64) - Type TINYTEXT - Q (dB) Minimum DOUBLE - Q (dB) Maximum DOUBLE - Q (dB) Average DOUBLE - Q (dB) Std Dev DOUBLE - Link Fail Seconds - In DOUBLE - Physical Error Count - In DOUBLE - Frame Check Sequence Error Count - In DOUBLE - Link Fail Seconds - Out DOUBLE - Physical Error Count - Out DOUBLE - Frame Check Sequence Error Count - Out DOUBLE - Unavailable Seconds (UAS) DOUBLE - Number of Seconds in Bin - Tx Line Card DOUBLE - Number of Seconds in Bin - Rx Line Card DOUBLE

Grafana Stats portal

PM data are monitored using Grafana dashboards.



Grafana portal details

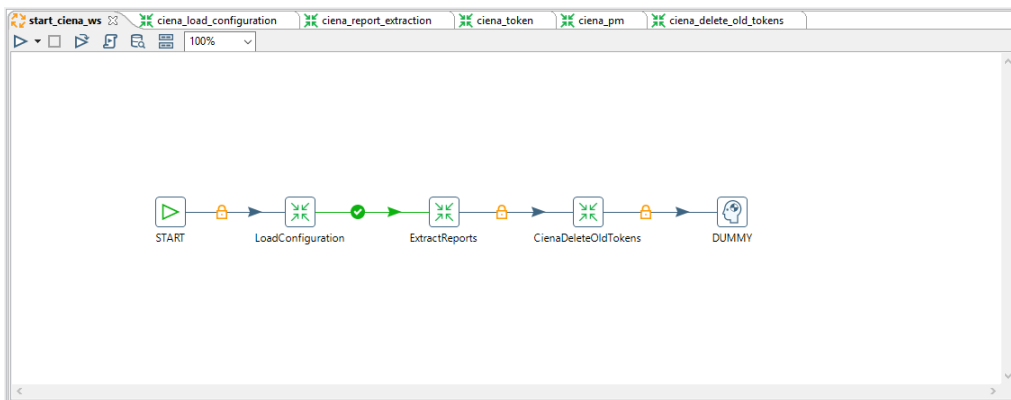
	Dashboard	URL	Git
Production - Customer	Customer name + " Service 3X Factors"	https://stats-local.local.hawaiki.co.nz/stats/d/kh215RwWk/customer-service-3x-factors?orgId=1	https://gitlab.local.hawaiki.co.nz/hawaiki-dev/kettle/tree/master/grafana
Production - NOC	Customer Service 3X Factors	https://stats-new.local.hawaiki.co.nz/stats/d/kh215RwWk/customer-service-3x-factors?orgId=1	
Pre-production	Customer Service 3X Factors	http://stats.test.local.hawaiki.co.nz/d/kh215RwWk/customer-service-3x-factors?orgId=1	

PM collection - Wave Server

Mediation application

Pentaho Data Integration(PDI) application collects PM data by calling REST API's to CIENA MCP server and extracts the Key Performance Index(KPI) values and stores it in the database(MariaDB).

PDI kettle job(start_ciena_ws.kjb) which starts the application



Environment details

	App Server	Application path	Ciena REST Server	Git	Collection time	Collection frequency
Production	10.7.2.104	/hawaiki_ii/ciena-ws	mcpcienza2.local.hawaiki.co.nz (AUSY)	https://gitlab.local.hawaiki.co.nz/hawaiki-dev/kettle/tree/master/ciena-ws	03, 17, 33, 47	15 minutes
Pre-production	10.10.12.108				03, 17, 33, 47	15 minutes

REST Request

Request body	KPI's collected	API
<pre>{ "data": { "attributes": { "criteria": { "binIndex": "1", "granularity": ["15_MINUTE"], "includeThreshold": false, "parameters": ["pcsunavailableSeconds", "pcseverelyErroredSeconds", "pcserroredSeconds", "txCrcErroredPackets", "txblock-errors", "txblock-errors"], "suppress": "NONE" }, "measurementPoints": [{ "modelId": "\$NCID", "resourceId": "\$NCID:TPE_\$NCID:EQPT_\$SHELF_\$SLOT-\$PORT-CTPServerToClient-\$SLOT-\$PORT", "resourceObject": { "location": { "port": "\$PORT", "shelf": "\$SHELF", "slot": "\$SLOT" }, "type": "pmData" } }] } } }</pre>	<i>pcserroredSeconds</i> - Errored Seconds - In, <i>txCrcErroredPackets</i> - Frame Check Sequence Error Count - In, <i>txCrcErroredPackets</i> - Frame Check Sequence Error Count - Out, <i>txblock-errors</i> - Physical Error Count - In, <i>txblock-errors</i> - Physical Error Count - Out , <i>pcseverelyErroredSeconds</i> - Severely Errored Seconds - In, <i>pcsunavailableSeconds</i> - Link Fail Seconds - In Note: we don't have "Link Fail Seconds - Out" in Wave server PM's	http://mcpcienza2.local.hawaiki.co.nz/pm/api/v1/queries

Request body	KPI's collected	API
<pre>"resourceType": "TPE" } } } }</pre>		

PM collection application details

File	Details
start_ciena_ws.kjb	This job handles everything from start to finish of the application. All other jobs and transformation are called from this job.
ciena_load_configuration.ktr	Loads configuration values from 'config.properties' file.
ciena_report_extraction.ktr	Handles getting access token and wave server pm extraction
ciena_token.ktr	Retrieves access token from Ciena MCP using REST API.
ciena_pm.ktr	Collects PM details from Ciena MCP using REST API and stores the data into the 'tems_och' table.
ciena_delete_old_tokens.ktr	Deletes old token from table ciena_tokens
config.properties	The configuration file contains the Ciena server and database details.
ciena_ws.log	Application log file.
ciena_ws.sh	A shell script that runs the PM collection application kettle job. This script is called from the mediation servers crontab.

Database

MariaDB details

	MariaDB Server	Schema	Tables used	Database backup
Production	ausy-db-mdb-oss-m2.local.hawaiki.co.nz	pm_db	ciena_tokens, service_source, ciena_ws_raw, tems_och	
Pre-production	10.10.13.201			

Tables

service_source - id INT(11) - customer VARCHAR(16) - customer_sid VARCHAR(16) - pm_source VARCHAR(128) - visible_to_customer TINYINT(4) - route_end VARCHAR(3) - segment_end VARCHAR(3) - source VARCHAR(16) - node_id VARCHAR(64) - to_from VARCHAR(2) - start_date DATETIME - end_date DATETIME	tems_och - id INT(10) - Date/Time DATETIME - customer_sid VARCHAR(24) - PM Source VARCHAR(256) - node_id VARCHAR(64) - Type TINYTEXT - Q (dB) Minimum DOUBLE - Q (dB) Maximum DOUBLE - Q (dB) Average DOUBLE - Q (dB) Std Dev DOUBLE - Link Fall Seconds - In DOUBLE - Physical Error Count - In DOUBLE - Frame Check Sequence Error Count - In DOUBLE - Link Fall Seconds - Out DOUBLE - Physical Error Count - Out DOUBLE - Frame Check Sequence Error Count - Out DOUBLE - Unavailable Seconds (JAS) DOUBLE - Number of Seconds in Bin - Tx Line Card DOUBLE - Number of Seconds in Bin - Rx Line Card DOUBLE	ciena_ws_raw - date_time DATETIME - granularityNativeSeconds INT(11) - cls_name TINYTEXT - ncid TINYTEXT - shelf TINYTEXT - slot TINYTEXT - port TINYTEXT - directionNative TINYTEXT - facilityNameNative TINYTEXT - parameterNative TINYTEXT - value DOUBLE	ciena_tokens - accessToken TINYTEXT - createdTime DATETIME - expirationTime DATETIME
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Grafana Stats portal

PM data are monitored using Grafana dashboards.



Grafana portal details

	Dashboard	URL	Git
Production - Customer	Customer name + " Service 8X Factors"	https://stats.local.hawaiki.co.nz/stats/d/MOM9mKik/amazon-service-8x-factors?orgId=1&var-customer=AMAZON&var-service_id=DEC-BC0067&var-pm_source=All	https://gitlab.local.hawaiki.co.nz/hawaiki-dev/kettle/tree/master/grafana

Production - NOC	Customer Service 8X Factors	https://stats-new.local.hawaiki.co.nz/stats/d/pVMmB4dmz/customer-service-8x-factors?orgId=1&var-customer=AMAZON&var-service_id=DEC-BC0067&var-pm_source=All
Production - NOC	Max Link Utilization	https://stats-new.local.hawaiki.co.nz/stats/d/afxMk9MRWk/max-link-utilization?orgId=1
Pre-production	Customer Service 8X Factors	http://stats.test.local.hawaiki.co.nz/d/pVMmB4dmz/customer-service-8x-factors?orgId=1&var-customer=AMAZON&var-service_id=DEC-BC0067&var-pm_source=All

PM collection - 5170

Mediation application

5170 PM collection application is written in Java using Spring Boot.

Environment details

	App Server	Application path	Ciena REST Server	Git	Collection time	Collection frequency
Production	10.7.2.104	/hawaiki_ill/five170	mcpcciena2.local.hawaiki.co.nz (AUSY)	https://gitlab.local.hawaiki.co.nz/hawaiki-dev/Five170	00, 15, 30, 45	15 minutes
Pre-production	10.10.12.108				00, 15, 30, 45	15 minutes

REST Request

Request body 1	KPI's collected	API
<pre>{ "data": { "attributes": { "filter": ["and", [{"key": "arg1"}, {"=", "facilityNameNative", "arg2"}]], "range": { "startTime": "arg3", "endTime": "arg4", "type": "absolute" } } } }</pre>	RX FCS error frames SLM frame loss ratio near end	http://mcpcciena2.local.hawaiki.co.nz/pm/api/v1/query/metrics

Request body 2	KPI's collected	API
<pre>{ "data": { "ncName": "arg1", "reportingFacility": "arg2", "granularity": "15-minute", "parameterNatives": [{"RX dropped frames", "RX accepted octets", "TX octets"}, {"TX octets"}], "range": { "type": "absolute", "startTime": "arg3", "endTime": "arg4" }, "synchronous": true } }</pre>	RX dropped frames RX accepted octets TX octets	http://mcpcciena2.local.hawaiki.co.nz/tdc/api/v2/diag/operMeas/facilities

PM collection application details

File	Details
Five170.jar	Executable jar file which is called by a cronjob.
application.properties	The configuration file contains the Ciena server, API, database details.
five170.log	Application log file.

Database

MariaDB details

	MariaDB Server	Schema	Tables used	Database backup
Production	ausy-db-mdb-oss-m2.local.hawaiki.co.nz	pm_db	five170_configuration, five170	
Pre-production	10.10.13.201			

Tables

five170_configuration - id INT(10) - customer VARCHAR(45) - source VARCHAR(45) - resource_id VARCHAR(45) - route_end VARCHAR(8) - source_1 VARCHAR(45) - source_2 VARCHAR(45) - port VARCHAR(45)	five170 - id INT(10) - source VARCHAR(16) - resource_id VARCHAR(45) - date_time DATETIME - slm_frame_loss_ratio_near_end DOUBLE - rx_fcs_error_frames DOUBLE - rx_dropped_frames DOUBLE - rx_accepted_octets DOUBLE - tx_octets DOUBLE
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Grafana Stats portal

PM data are monitored using Grafana dashboards.



Grafana portal details

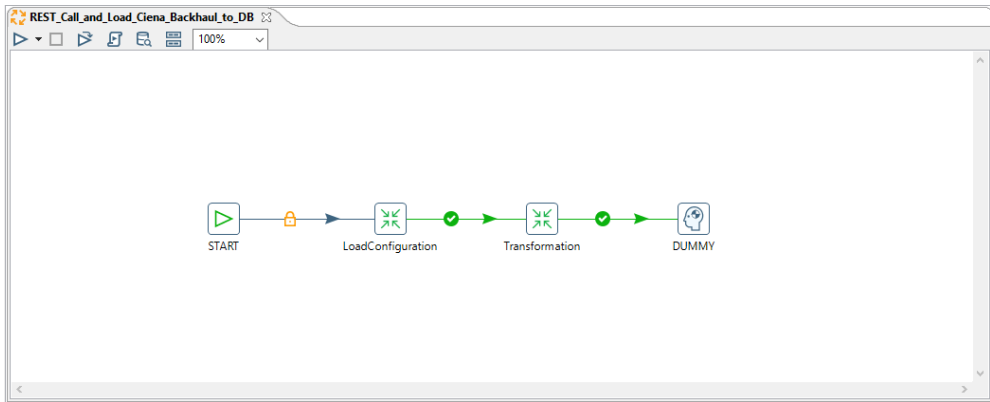
	Dashboard	URL	Git
Production - Customer	Customer name + " Five170"	https://stats.local.hawaiki.co.nz/stats/d/4D8T_YpWk/aussie-broadband-five170?orgId=1	https://gitlab.local.hawaiki.co.nz/hawaiki-dev/kettle/tree/master/grafana
Production - NOC	Five170	https://stats-new.local.hawaiki.co.nz/stats/d/uoDL_LtZz/five170?orgId=1	
Production - NOC	Five170 - Max Link Utilization	https://stats-new.local.hawaiki.co.nz/stats/d/dy7t3LWk/five170-max-link-utilization?orgId=1	
Pre-production	Five170	http://stats.test.local.hawaiki.co.nz/d/uoDL_LtZz/five170?orgId=1&from=now-7d&to=now	

Amplifiers Power (Ciena backhaul)

Mediation application

Pentaho Data Integration(PDI) application collects backhaul data by calling REST API's to CIENA MCP server and extracts the Key Performance Index(KPI) values and stores it in the database(MariaDB).

PDI kettle job(REST_Call_and_Load_Ciena_Backhaul_to_DB.kjb) which starts the application



Environment details

	App Server	Application path	Ciena REST Server	Git	Collection time	Collection frequency
Production	10.7.2.104	/hawaiki_ill/ciena	mcpcienza2.local.hawaiki.co.nz (AUSY)	https://gitlab.local.hawaiki.co.nz/hawaiki-dev/kettle/tree/master/ciena	06, 21, 35, 50	15 minutes
Pre-production	10.10.12.108				05, 20, 35, 50	15 minutes

REST Request

Request body	KPI's collected	API
<pre>{ "data": { "attributes": { "criteria": { "binIndex": "1", "granularity": ["15_MINUTE"], "includeThreshold": false, "parameters": { "OP\$IN_OUTMIN-OTS", "OP\$IN_OUTMAX-OTS", "OP\$IN_OUTAVG-OTS" }, "suppress": "NONE" }, "measurementPoints": [{ "nodeId": "\$NCID", "resourceId": "\$NCID::TPE_\$NCID::EQPT_\$SHELF_\$SLOT_\$PORT-</pre>	USPA - Transmit, USPA - Received, USFB - Transmit, USFB - Received	http://mcpcienza2.local.hawaiki.co.nz/pm/api/v1/queries

Request body	KPI's collected	API
<pre>PTP", "resourceObject": { "location": { "port": "\$PORT", "shelf": "\$SHELF", "slot": "\$SLOT" }, "type": "pmData" }, "resourceType": "TPE" } } }</pre>		

PM collection application details

File	Details
REST_Call_and_Load_Ciena_Backhaul_to_DB.kjb	This job handles everything from start to finish of the application. All other jobs and transformation are called from this job.
ciena_load_configuration.ktr	Loads configuration values from 'config.properties' file.
ciena_backhaul_report_extraction.ktr	Handles getting access token and backhaul data extraction
ciena_token.ktr	Retrieves access token from Ciena MCP using REST API.
ciena_backhaul_pm.ktr	Collects backhaul PM details from Ciena MCP using REST API and stores the data into the 'ciena_backhaul_raw' table.
config.properties	The configuration file contains the Ciena server and database details.
ciena_backhaul.log	Application log file.
ciena_backhaul.sh	A shell script that runs the PM collection application kettle job. This script is called from the mediation servers crontab.

Database

MariaDB details

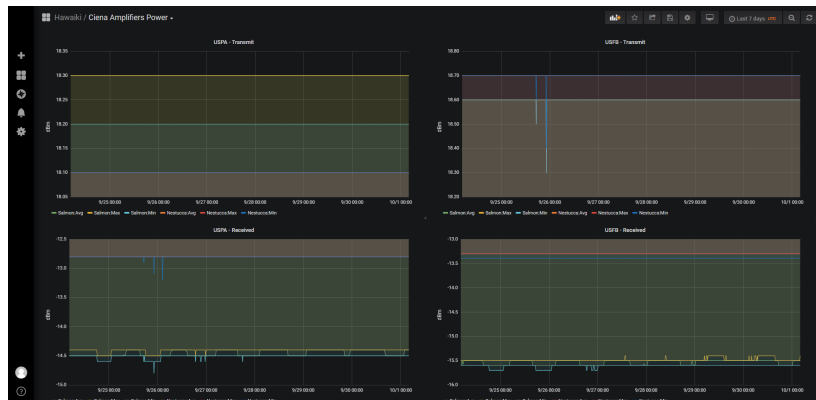
	MariaDB Server	Schema	Tables used	Database backup
Production	ausy-db-mdb-oss-m2.local.hawaiki.co.nz	pm_db	ciena_tokens, ciena_backhaul_raw	
Pre-production	10.10.13.201			

Tables

ciena_backhaul_raw - date_time DATETIME - granularityNativeSeconds INT(11) - cls_name TINYTEXT - ncid TINYTEXT - shelf TINYTEXT - slot TINYTEXT - port TINYTEXT - directionNative TINYTEXT - facilityNameNative TINYTEXT - parameterNative TINYTEXT - value DOUBLE	ciena_tokens - accessToken TINYTEXT - createdTime DATETIME - expirationTime DATETIME
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Grafana Stats portal

Backhaul data are monitored using Grafana dashboards.



Grafana portal details

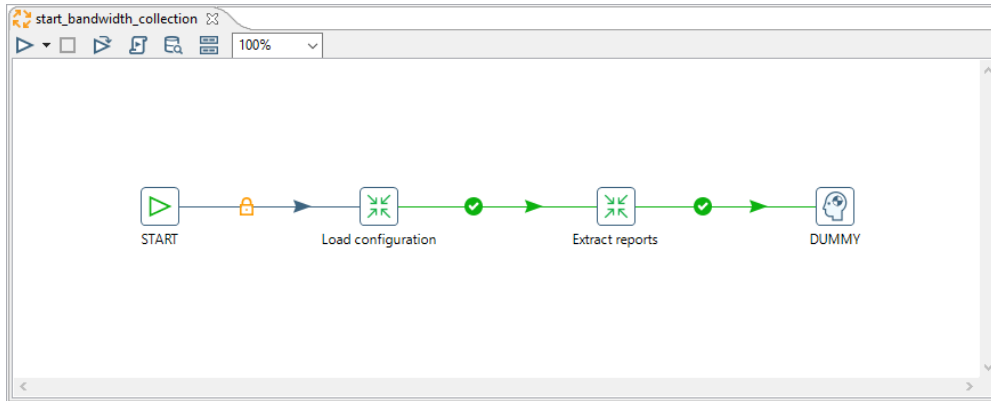
	Dashboard	URL	Git
Production - NOC	Ciena Amplifiers Power	https://stats-new.local.hawaiki.co.nz/stats/d/PJfy5vcmk/ciena-amplifiers-power?orgId=1	https://gitlab.local.hawaiki.co.nz/hawaiki-dev/kettle/tree/master/grafana
Pre-production	Ciena Amplifiers Power	http://stats.test.local.hawaiki.co.nz/d/PJfy5vcmk/ciena-amplifiers-power?	

Bandwidth Collection

Mediation application

Pentaho Data Integration(PDI) application collects bandwidth data by calling REST API's to CIENA MCP server and extracts the Key Performance Index(KPI) values and stores it in the database(MariaDB). As of September 2019, this application collects bandwidth data only for Wave Server services.

PDI kettle job(start_bandwidth_collection.kjb) which starts the application.



Environment details

	App Server	Application path	Ciena REST Server	Git	Collection time	Collection frequency
Production	10.7.2.104	/hawaiki_ill/bandwidth	mcpciena2.local.hawaiki.co.nz (AUSY)	https://gitlab.local.hawaiki.co.nz/hawaiki-dev/kettle/tree/master/bandwidth	02, 16, 32, 46	15 minutes
Pre-production	10.10.12.108				02, 16, 32, 46	15 minutes

REST Request

Request body	KPI's collected	API
<pre>{ "data": { "attributes": { "criteria": { "binIndex": "1", "granularity": ["15_MINUTE"], "includeThreshold": false, "suppress": "NONE" } }, "measurementPoints": [{ "model": "SNCID", "resourceId": "SNCID:TPE_SNCID:EQPT_SHELF_SLOT-\$PORT-CTPServerToClient-\$SLOT-\$PORT", "resourceObject": { "location": { "port": "\$PORT", "shelf": "\$SHELF", "slot": "\$SLOT" }, "type": "pmData" }, "resourceType": "TPE" }] } }</pre>	rxminvalue, rxmaxvalue, rxavgvalue, txminvalue, txmaxvalue, txavgvalue	http://mcpciena2.local.hawaiki.co.nz/pm/api/v1/queries

Bandwidth collection application details

File	Details
start_bandwidth_collection.kjb	This job handles everything from start to finish of the application. All other jobs and transformation are called from this job.
bandwidth_load_configuration.ktr	Loads configuration values from 'config.properties' file.
bandwidth_report_extraction.ktr	Handles getting access token and bandwidth extraction.
get_token.ktr	Retrieves access token from Ciena MCP using REST API.
get_pms.ktr	Collects bandwidth data from Ciena MCP using REST API and stores the data into the 'bandwidth_data_collection' table.
config.properties	The configuration file contains the Ciena server and database details.
bandwidth.log	Application log file.
bandwidth.sh	A shell script that runs the bandwidth collection application kettle job. This script is called from the mediation servers crontab.

Database

MariaDB details

	MariaDB Server	Schema	Tables used	Database
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				backup
Production	ausy-db-mdb-oss-m2.local.hawaiki.co.nz	pm_db	configuration, bandwidth_data_collection	
Pre-production	10.10.13.201			

Tables

bandwidth_data_collection	configuration
id INT(10)	id INT(11)
collectiontime TIMESTAMPTZ	serviceid VARCHAR(45)
service_id VARCHAR(45)	sourcename VARCHAR(45)
node_id VARCHAR(45)	nodeid VARCHAR(45)
sourcename VARCHAR(45)	vendor VARCHAR(45)
rxmavalue DOUBLE	collection_duration INT(11)
txmavalue DOUBLE	route_direction VARCHAR(45)
rxminvalue DOUBLE	segment_direction VARCHAR(45)
txminvalue DOUBLE	nodesplayname VARCHAR(45)
rxavgvalue DOUBLE	disable_collection TINYINT(4)
txavgvalue DOUBLE	visible_to_customer TINYINT(4)

Grafana Stats portal

Bandwidth data are monitored using Grafana dashboards and usages are shown in directional, eg: South to North.

The endpoints of the routes and segments are managed by the fields `route_direction`, `segment_direction` in the `configuration` table.

The service starting point of the south is marked as 'A' end, and the north endpoint is marked as 'Z' end in the `route_direction` field in the `configuration` table.

Endpoints of each segment in the route are marked as 1A, 1Z, 2A, 2Z order from south to north in the `segment_direction` field in the `configuration` table.

Bandwidth Utilization Directional - Route-based



Max Link Utilization



Grafana portal details

	Dashboard	URL	Git
Production - NOC	Bandwidth Utilization Directional - Route based	https://stats-new.local.hawaiki.co.nz/stats/d/afxMk9MRWk/bandwidth-utilization-directional-route-based?orgId=1&from=now-1M%2F&to=now-1M%2F	https://gitlab.local.hawaiki.co.nz/hawaiki-dev/kettle/tree/master/grafana
Production - NOC	Max Link Utilization	https://stats-new.local.hawaiki.co.nz/stats/d/afxMk9MRWk/max-link-utilization?orgId=1&var-customer=AMAZON&var-service_id=DEC-BC0066&var-pm_source=AUSY:PORT-2-7	
Pre-production	Bandwidth Utilization Directional - Route based	http://stats.test.local.hawaiki.co.nz/d/afxMk9MRWk/bandwidth-utilization-directional-route-based?orgId=1&from=now-1M%2F&to=now-1M%2F	
Pre-production	Max Link Utilization	http://stats.test.local.hawaiki.co.nz/d/afxMk9MRWk/max-link-utilization?orgId=1&from=now-1M%2F&to=now-1M%2F&var-customer=AMAZON&var-service_id=DEC-BC0066&var-pm_source=AUSY:PORT-2-7	

Alarm collection - Bash Script Application

Ciena bash script application collects alarms through Ciena MCP REST API calls.

Mediation application

Alarms received are cross-checked with customer assigned ports and sends to OpenNMS.
All the alarms which are assigned to a customer are forwarded to Impacted Services in OpenNMS.
All the alarms which are not assigned to a customer are forwarded to Ciena Alarm in OpenNMS.

Environment details

	App Server	Application path	Ciena MCP Server	Git	Collection frequency
Production	10.7.1.41	/opt/HawaikiTools/CienaAlarms	mcpciena2.local.hawaiki.co.nz (AUSY)	https://gitlab.local.hawaiki.co.nz/hawaiki-dev/kettle/tree/master/ciena-alarms	1 minute

Ciena alarm collection application details

File	Details
CienaAlarms.sh	Bash script which contains application code.
CienaCustomerPMs.csv	File which contains customer assigned port details. (This file has to be updated when a new service is added for PM collection)
CienaAlarmsChecksum.csv	A temporary file used for filtering duplicate alarms.
CienaAlarmsClearedChecksum.csv	The temporary files used by the script.
Ciena_Alarms_Cleared.txt	
Ciena_Alarms_Cleared.csv	
Ciena_Alarms.txt	
Ciena_Alarms.csv	

Email notifications

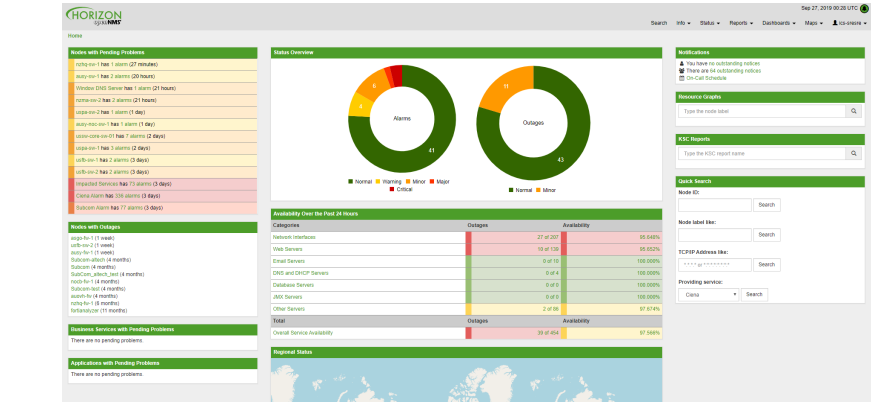
Condition	to address
alarms match customer assigned port	itdev@hawaikicable.co.nz, pat.curtis@hawaiki.co.nz, thasa.b@hawaikicable.co.nz, operations-noc@hawaikicable.co.nz

Database

Ciena alarms are not stored in the database by bash script mediation application.
OpenNMS application stores it in PostgreSQL.

OpenNMS

Ciena alarms are monitored using OpenNMS



OpenNMS portal details

	URL
Production	https://opennms.local.hawaiki.co.nz/opennms/index.jsp

Alarm collection - Java Application

Ciena java application collects alarms through Ciena MCP REST API calls.
As of September 2019 Alarm collected using Java application is not used anywhere, just storing all the alarms in MariaDB.

Mediation application

Environment details

	App Server	Application path	Ciena MCP Server	Git	Collection frequency
Pre-production	10.10.12.108	/hawaiki_il/alarms/ciena	mcpciena2.local.hawaiki.co.nz (AUSY)	https://gitlab.local.hawaiki.co.nz/sreeraj.sreenivasan/alarms	1 minute

Ciena alarm collection java application details

File	Details
alarm.jar	Executable jar file which is called by a cronjob.
application.properties	The configuration file contains the Ciena MCP server, API, database details.
alarm.log	Application log file.

Database

MariaDB details

	MariaDB Server	Schema	Tables used	Database backup
Pre-production	10.10.13.201	alarm	raw_alarm_java	

Table

raw_alarm_java
id INT(11)
time_inserted TIMESTAMP
type TINYTEXT
_id TINYTEXT
alarm_id TINYTEXT
node_id TINYTEXT
ra_alarm_id TINYTEXT
node_type TINYTEXT
state TINYTEXT
resource TINYTEXT
native_condition_type TINYTEXT
condition_severity TINYTEXT
service_affecting TINYTEXT
manual_clearable TINYINT(4)
additional_text TINYTEXT
first_raise_time TINYTEXT
last_raise_time TINYTEXT
clear_time TINYTEXT
number_of_occurrences INT(11)
acknowledge_state TINYTEXT
device_id TINYTEXT
device_name TINYTEXT
ip_address TINYTEXT
mac_address TINYTEXT
sequence_id TINYTEXT

Fault Tolerance In Mediation Layer

[Fault Tolerance In Mediation Layer](#)